

Oluwasegun Adewola

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EDUCATION

Northeastern University

Expected Dec. 2027

*Master of Science in **Computer Science**, GPA: 4.0*

Seattle, USA

- Relevant Coursework: Distributed Systems, Machine Learning, Databases Management Systems, Object Oriented Design, Data Structures and Algorithms, Advanced Algorithms, Discrete Math
- People's Choice **Winner**, **Qualcomm & Microsoft** Edge AI Hackathon
- **Global Learner Scholarship**

EXPERIENCE

Graduate Teaching Assistant

Aug. 2025 – Apr. 2026

Khoury College of Computer Science, Northeastern University

Seattle, USA

- Led **Discrete Math** and **Data Structures and Algorithms** instruction across **2** semesters (Fall 2025, Spring 2026) through teaching, office hours, and grading under Jelani Drew and Prof. Tony Mullen.

Software Engineer

Dec. 2022 – Aug. 2024

*Flutterwave **YC16** (Digital Payments for African businesses)*

Lagos, Nigeria

- Drove a **66%** revenue increase in the first month by leading development of the predictive pricing service for **SendApp**, Flutterwave's consumer currency exchange product, covering **660+** currency pairs, built in **Python** and deployed via **GitHub Actions**.
- Designed and rolled out initial dynamic pricing algorithms validated by domain experts using **Linear Programming**, and built **Power BI** monitoring dashboards enabling real-time error detection and alerting, saving **\$1.5M+** in revenue over 6 months.
- Built scheduled web crawlers and data pipelines using **Requests**, **Playwright**, and **Google Cloud** to aggregate competitor exchange rates on a recurring schedule, powering both the predictive pricing service and executive market intelligence reporting.

PROJECTS

Distributed Chat App | Java, Spring Boot, RabbitMQ, Cassandra, Redis, AWS (ALB, EC2)

Jan. 2026 – Apr. 2026

- Built a distributed chat system with **4 Spring Boot WebSocket** servers behind a load balancer, **RabbitMQ** for message routing, **Redis** for caching and **Cassandra** for storage, with dedicated consumers decoupling message delivery from persistence, all on self-managed **AWS EC2** instances.
- Achieved **13K messages/second** throughput validated through load and integration testing, with dead letter queues, exponential backoff, and circuit breaker patterns for production-grade fault tolerance.

NeuPool (Carpool Co-ordination App) | Go, TypeScript, React, PostgreSQL, SQS, Redis, AWS

Jan. 2026 – Ongoing

- Carpooling coordination platform for universities and large organisations, currently piloting at **Northeastern University** and targeting wide release in Fall 2026, with **RBAC** group management, **WebSocket**-based in-app chat, push notifications, and gas cost splitting via **Stripe**.
- Built with a **Go** backend and **TypeScript/React** frontend on **AWS (EC2, RDS, SQS, ElastiCache)**, using **Go's native concurrency (goroutines)** to handle **1,000+** simultaneous matching requests, **Google OR-Tools** for optimal ride assignment, and **Amazon SQS** to decouple event processing.

RemitTrack (Exchange Rate Comparison App) | Go, React, PostgreSQL, Kinesis, Redis, AWS

Jun. 2025 – Ongoing

- Built a real-time exchange rate comparison platform for the African markets covering **15+** Money Transfer Providers, with user price-target alerts and an **LLM**-powered chat interface for natural-language rate queries as an accessibility feature.
- Architected a fully **Go** backend and **TypeScript/React** frontend on **AWS (EC2, RDS for PostgreSQL, Kinesis, ElastiCache for Redis)** with a **goroutine**-based **ETL** pipeline for concurrent provider ingestion; cached rate data ensures low-latency reads, aggregated data doubles as a market intelligence and arbitrage risk signal for MTP operators, with integration tests covering scraper correctness and rate normalization.

SKILLS

Languages: Go, Java, Python, C++, C, JavaScript, TypeScript, Bash, SQL, HTML/CSS

Technologies: Spring/Spring Boot, FastAPI, Node.js, React.js, RabbitMQ, Kafka, TCP/IP

Databases/Storage: PostgreSQL, MySQL, Redis, Cassandra, MongoDB, DynamoDB, S3

Tools: Linux, AWS, GCP, Docker, Kubernetes, Git, Github Actions, Terraform, Prometheus, Grafana

AI/ML: LangChain, PyTorch, scikit-learn, NumPy

LEADERSHIP & PARTICIPATION

Seattle Systems (seattlesystems.xyz) – *Member*

Jan. 2026 – Present

Northeastern University Leadership Club – *Co-President*

Jan. 2025 – Present

National Society of Black Engineers (NSBE) – *Member*

Sep. 2024 – Present